Azure Data Engineer Interview Questions (Scenario-Based - 2026)

Round 1

AZURE DATA FACTORY (ADF) SCENARIOS

1. Your ADF pipeline fails due to a missing file in the source storage. How do you handle this failure automatically?
2. You need to copy data from on-prem SQL Server to Azure SQL Database every day. How will you design this pipeline securely and efficiently?
3. You have 100 CSV files daily with different names but the same schema. How can you load all of them dynamically in one pipeline?
4. One of your Copy Data activities takes too long to complete. How will you troubleshoot and optimize it?
5. You need to pass parameters such as file name and date between pipelines. How will you do that?
6. You must trigger a pipeline only when a new file arrives in Azure Blob Storage. How will you implement it?
7. Your pipeline must load data into multiple tables from one file. How do you design it?
8. You want Pipeline B to run only after Pipeline A completes successfully. How do you achieve dependency?
9. You must run a pipeline hourly but only between 9 AM and 6 PM. How do you configure it?
10. You want to perform transformations inside ADF before loading into SQL. Which activity would you use?

AZURE DATA LAKE & STORAGE SCENARIOS

11. You need to store both raw and curated data versions in your lake. How do you design your Data Lake zones?
12. Your team complains that data engineers are overwriting production files. How will you prevent accidental overwrites?
13. You want to read JSON and Parquet files together in the same folder. What's the best approach?
14. You have millions of small files causing performance issues. How do you fix it?

AZURE SQL / SYNAPSE SCENARIOS

15. You're loading 10 million rows daily into Azure SQL Database and it's slow. How can you speed it up?
16. Your query performance is poor in Synapse Dedicated SQL Pool. How will you optimize it?
17. You need to transform data before loading into Synapse. Would you use ADF Data Flow or Synapse pipeline, and why?

18. You must perform incremental load into Synapse. How would you handle it?
19. You need to merge data (update if exists, insert if not). How would you achieve this in Azure?

AZURE DATABRICKS SCENARIOS

20. You have 100 GB of CSV data to clean daily using Databricks. How do you optimize Spark performance?
21. Your Spark job failed due to memory issues. How do you troubleshoot it?
22. You must handle duplicates in Databricks. How would you remove them efficiently?
23. You want to maintain history of changes (SCD Type 2) in Databricks Delta. How do you do it?
24. You're asked to share processed data from Databricks with Power BI. How do you connect them?

MONITORING, ALERTS, AND OPTIMIZATION SCENARIOS

25. Your ADF pipeline randomly fails once every few days. How do you debug the root cause?
26. You need to send an email alert when a pipeline fails. How do you do that in Azure?
27. Your team wants to track data-load status (Success/Failure/Duration) daily. How do you implement it?
28. You need to run five pipelines in parallel but limit concurrency to two. How do you manage it?
29. Your ADF pipeline should automatically retry failed activities three times. How do you configure that?
30. Your Synapse query cost is too high. How do you optimize costs?

REAL-TIME SCENARIOS

31. You must process IoT sensor data in real time. Which Azure components will you use?
32. Your streaming job is delayed. How do you fix latency issues?
33. You must handle late-arriving events in Stream Analytics. How do you ensure correct aggregation?

GENERAL PROJECT / DESIGN SCENARIOS

34. Design an end-to-end Azure Data Pipeline for a retail company (CSV → Data Lake → SQL → Power BI). Which services would you use and why?
35. How do you ensure data quality in your Azure pipelines?
36. How do you version-control your ADF pipelines?
37. How do you promote pipelines from Dev to Test to Prod?
38. How do you secure credentials in your pipelines?
39. Your client asks for a daily data-load report automatically. How would you deliver it?
40. How would you estimate cost for your ADF pipeline?